Fiscal Year:	FY 2013	**Task Last Updated:**	FY 01/02/2013
PI Name:	Park, Sung-Yong Ph.D.		
Project Title:	Heat Transfer on Multiscale Textures with Electrowetting Control		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Thousand Oaks	**State:**	CA
Zip Code:	91360-2362	**Congressional District:**	24
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
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No. of Post Docs:		**No. of PhD Degrees:**	3
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/2012; okayed per D. Chao/GRC (Ed., 8/30/12)		
Key Personnel Changes/Previous PI:	Sung-Yong Park is the new PI effective in FY2012. Previous PI Dr. Chung-Lung Chen retired from Teledyne in November 2011.		
COI Name (Institution):			
Grant/Contract No.:	NNC09CA36C		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: PI change for FY2012. See Chung-Lung Chen (PI) and project titled "Heat Transfer on Multiscale Textures with Electrowetting Control" for previous reports--FY2009-FY2011. Modern electronic systems used in space must be reliable and efficient with thermal management unaffected by outer space environments. Heat pipes and vapor chambers based on phase-change heat transfer mechanisms have proven to be an efficient approach to dissipate heat in many space or ground-based thermal-energy systems. However, a layer of capillary wick structure fully covering the chamber walls, e.g., condenser and evaporator surfaces, creates additional thermal resistance, because the effective thermal conductivity is much lower in saturated wicks. This thermal barrier can generate an unacceptable temperature rise, especially in high heat-flux applications.		

Task Description:	To overcome this issue, we propose to develop an innovative wickless and electrowetting-controlled vapor chamber with bioinspired two-tier textures on the chamber wall. These engineered superhydrophobic / superhydrophilic multiscale textures, formed by growing carbon nanotubes (CNTs) on top of micropillars on a silicon chip, mimic the structure of lotus leaves; they can promote dropwise condensation at the condenser site, and enhance film evaporation at the evaporator site. The condensate droplets will be dislodged from the condenser site and circulated to the heat source via electrowetting transport. Droplet transport by electrowetting without any mechanical moving parts has great potential in thermal management due to its programmable and reconfigurable characteristics, prompt response, and low power consumption. Most importantly, given a two-dimensional array of control electrodes, condensate can be transported over the low-friction hydrophobic surface to a hot spot without the significant pressure loss that is associated with fine pore wicks. The transported droplets will be captured by a few finger-like extensions from the heat source, which are hydrophilized by a coating of UV&Ozone-treated carbon nanotubes. This hydrophilic coating will force droplets to cling to the hot spot surface, changing heat transfer into the more efficient form of thin film evaporation. In this program, we aim to a) investigate wetting characteristics and phase-change heat transfer on multiscale textures; b) quantify electrowetting transport of condensates and transient heat transfer on the textured surface; c) develop numerical models of heat transfer on the textures. Reduced gravity flight tests will be performed to validate our analysis and ground tests. This innovative design from this program will enable a new generation of high-performance space integrated systems to operate at high-power densities without problems from temperature gradients, increased weight, or added complexity. Low-pressure-loss droplet transport on multiscale textures, combined with adaptable, low-power electrowetting, has the potential to significantly improve vapor chamber efficiency, increase its robustness and lifetime, and reduces the mass of space flight systems needed to support future human space exploration activities. The application of nanothermal materials and adaptive liquid transport technology based on electrowetting will lead to a major breakthrough in the thermal management field. The success of the proposed effort can considerably enhance thermal management in the future space explorations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	One of the major applications of phase-change heat transfer at low-gravity conditions is the design and maintenance of robust and efficient thermal control systems for future space exploration and satellite applications. Heat pipes and vapor chambers based on phase-change heat transfer mechanisms have proven to be good candidates to dissipate heat in many space thermal-energy systems. However, a layer of capillary wick structure fully covering the heat pipe and vapor chamber inner walls creates additional thermal resistance because the effective thermal conductivity is much lower in saturated wicks. This thermal barrier can generate an unacceptable temperature rise, especially in high heat-flux and harsh-environment applications. To address this issue, Teledyne Scientific & Imaging Company (TS&I) proposes to develop an innovative wickless and electrowetting-controlled vapor chamber with bioinspired two-tier textures on the chamber wall. This innovative design will enable a new generation of high-performance space integrated systems to operate at high-power densities without problems from temperature gradients, increased weight, or added complexity. Rapid droplet transport on multiscale textures, combined with adaptable, low-power electrowetting, has the potential to significantly improve vapor chamber efficiency, increase its robustness and lifetime, and reduce the mass of space flight systems needed to support future human space exploration activities. The application of nanothermal materials and adaptive liquid transport technology based on electrowetting will lead to a major breakthrough in the thermal management field. The success of the proposed effort can considerably enhance thermal management in the future space explorations.
	Recently electrowetting-on-dielectric (EWOD) has emerged as one of the effective on-chip cooling techniques. It can provide several advantages over other active-type thermal management technologies, including (1) extremely low power consumption for fluid transport in the range of ~μW; (2) no mechanical pump or valve required; (3) no need of high pressure head encountered in continuous flows confined in microchannels, and (4) fast response time. Most importantly, EWOD allows rapid droplet transportation along any arbitrary path to the target heat sources where heat flux concentrated can be effectively controlled. Conventional EWOD devices are typically implemented by sandwiching droplets between two parallel plates and droplet actuation is achieved by addressing digitized electrodes. For on-chip thermal management using EWOD, a coolant liquid droplet is delivered to the target heating locations and heat generated from the hot spots can be transferred and released by the droplets. This procedure is repeated by next droplets for continuous cooling. Such EWOD-based cooling performance can be further improved by using liquid metals or alloys that have a few orders of magnitude higher thermal conductivity than water's. Since a sandwiched EWOD configuration does not allow vaporization of coolant droplets, all previously demonstrated EWOD-based cooling has been achieved by only relying on sensible heat transfer, which results in very limited heat rejection of the device. To improve thermal rejection capability, we have proposed to develop a single-sided EWOD (SEWOD) technology that enables two-phase heat transfer on a single-sided hot spot surface. Thus thermal rejection capability can be significantly enhanced. We have developed the new electrical control system utilizing National Instrument's LabVIEW software to control the timing and regulation of the excitation voltage which was used for determining the droplet volume and dispensing frequency. With the LabVIEW-based pair control scheme, we have demonstrated a full set of droplet manipulations necessary for effective cooling on SEWOD, including continuous droplet injection with precise volume control than 2% variation, rapid transport at a speed as fast as 7.5 cm/s, and droplet immobilization on the target hot spots. All previously demonstrated heat transfer studies using EWOD have been conducted only on the hydrophobic hot spot surface. It is worthwhile to investigate surface effect on cooling of the SEWOD. For this study, we prepared two SEWOD devices with different hot spot surface properties: one was a hydrophobic surface provided by spin coating of a FluoroPel polymer solution and another was a hydrophilic surface on a thin Ta2O5 film deposited by an e-beam evaporator. The surface properties were examined by measuring its contact angle. The droplet sits on the hydrophobic surface with the contact angle of 118°, while the angle was measured to be 68° on the hydrophilic Ta2O5 coated surface. Our cooling tests indicate that the hydrophilic surface property was able to effectively remove heat from the hot spot. On the hydrophobic hot spot surface, a droplet was rarely positioned on the hot spot area. Due to low friction or less pinning effect on the hydrophobic surface, it just skittered away and thus the hot spot was partially covered by the coolant

Task Progress:	droplet. During evaporation process, explosive vapor steam drives the droplet away from the target hot spots, resulting in long evaporation period and quick temperature increase. This cooling study was repeated on the hydrophilic hot spot surface to see how the hot spot surface property has an influence on heat transfer. The delivered droplet was firmly stuck and expanded the contact area on the hydrophilic hot spot surface. The temperature instantly and greatly dropped down from 165°C (initial hot spot temp.) to 102°C. The hot spot was maintained around 100°C until the droplet evaporation is fully developed. One interesting observation is the shorter duration of droplet vaporization on the hydrophilic surface. This indicates that the heat dissipation rate is much faster on the hydrophilic surface than the hydrophobic. Our studies conclude that the hydrophilic surface modification on the target heat sources can be very useful to improve heat transfer capability on SEWOD. Not only can a hydrophilic surface make a droplet firmly stick onto the hot spot surface even during explosive vaporization, but also increase the heat transfer area by expanding the droplet contact area. With SEWOD's capabilities of various droplet manipulation and hydrophilic hot spot surface, we have demonstrated continuous cooling using four successive droplets. A hot spot was initially heated up to 172°C under the heat flux density of 6.9 W/cm2. When a 30 µL water droplet firstly arrived at the hydrophilic hot spot, the temperature instantly dropped down to 89.5°C within 2s and was maintained less than 100°C for 106s. At the moment of completing the droplet vaporization, the next droplet was quickly delivered to avoid overheating. This process was continued and the hot spot temperature was able to be mitigated below 100°C for 475s using four successive droplets. Further experiment was conducted to study high heat flux control. LabView control scheme allows slowly increasing the heater power with the rate of 0.1V/4s and continuously delivering the 6.7 µL droplets. Upon arrival of the droplets, they spread out and increased the contact area on the hydrophobic-treated hot spot, which experienced an instant temperature drop around 6 ~ 7°C, and then the temperature was recovered. These processes were repeated. Around the heat flux of 20 W//cm2, we observed that bubbles began to initiate at nucleation sites and were slowly escaped. This nucleate boiling was gradually increased as increasing the heat flux. To mitigate the hot spot temperature, the frequency of droplet supply was further increased. When droplets were delivered to the target hot locations, the surface temperature was instantly cooled down and quickly recovered with the temperature fluctuation around 6 ~ 10°C before the heat flux reached 50 W//cm2. Multiple nucleation bubbles generation and escaping were observed, which enhances internal convection flow inside the droplet on the hot spot. Sometimes, explosive vaporization during bubbles evacuation caused to pop up and move away small parts of water from the hot spot location. At the high heat flux above 50 W/cm2, the nucleate boiling occurs very intensively, while coolant droplets were supplied even more. Much more nucleation sites became active and increased bubble formation caused bubble interference, coalescence and release. We observed that droplets were supplied even faster than vaporization, which caused droplet accumulation around the hot spot. This thick layer of water increases thermal resistance and limits cooling performance of the devices. Although the heat flux was increased up to 131 W/cm2, the superheat temperature was maintained below 41°C. We stopped the experiments to avoid the device burnout. By further developing a feedback control scheme that enables to control droplet volume and feeding speed to mitigate the hot spot temperature, this novel SEWOD-driven cooling technique promises to potentially function as a wickless vapor chamber or heat pipe with enhanced thermal managing capabilities. In the third phase of the program, we have participated in reduced gravity flight test. Since parabolic maneuver of the airplane could provide a zero gravity condition only for 20 seconds, we decided to run simple micro-scale fluidic experiments. We conducted two separate experiments for studying: (1) gravity effect on a water sessile droplet on a hydrophilic surface and (2) SEWOD-driven droplet dynamics and its gravity effect. Water droplet with three different volumes (i.e. three different Bond number) were prepared and we observed their shape change under zero gravity. These results were compared with the ones in normal gravity. For the droplet with Bo>1, the spherical droplet shape was still maintained under zero gravity, while its shape was suppressed in normal gravity. Another study was about SEWOD-driven droplet actuation and dynamics affected by gravity. For this study, the device was tilted to provide a gravitation force that pulls down against the droplet driving up. We have observed that, under zero gravity, a 60µL droplet can be driven at the sloped angle of 22.5° with a given voltage of 50V, which never happen in normal gravity environment.
Bibliography Type:	Description: (Last Updated: 09/09/2022)
Papers from Meeting Proceedings	Park S-Y, Cheng J, Chen C-L. "Active Hot Spot Cooling Controlled by Single-sided Electrowetting-on-Dielectric (SEWOD)." ASME 2012 Summer Heat Transfer Conference (ASME HT 2012), Puerto Rico, USA, July 8-12, 2012. ASME 2012 Summer Heat Transfer Conference (ASME HT 2012), Puerto Rico, USA, July 8-12, 2012. Paper HT2012-58111. https://doi.org/10.1115/HT2012-58111 , Jul-2012